

Question	Answer	Marks
6(a)(i)	$R = V / I$	C1
	resistance = $(12 / 0.20) / 2$ or $6 / 0.20$ $= 30 \Omega$	A1
6(a)(ii)	$I = 0.50 - 0.20$ (= 0.30 A)	C1
	$R + 28 = 12 / 0.30$ (= 40 Ω) $R = 12 \Omega$	A1

Question	Answer	Marks
6(b)	p.d. across lamp = 0.20×30 (= 6.0 V)	C1
	p.d. across $R = 0.30 \times 12$ (= 3.6 V)	C1
	$V_{XY} = 6.0 - 3.6$ = 2.4 V	A1
	or	
	p.d. across lamp = 0.20×30 (= 6.0 V)	(C1)
	p.d. across 28Ω resistor = 0.30×28 (= 8.4 V)	(C1)
	$V_{XY} = 8.4 - 6.0$ = 2.4 V	(A1)
6(c)	$P = VI$ or $P = EI$ or $P = I^2R$ or $P = V^2/R$	C1
	ratio = $(6.0 \times 0.20) \times 2 / (12 \times 0.50)$ or $0.20 / 0.50$ = 0.40	A1
6(d)	no change to V across lamps, so power in lamps unchanged or current in battery/total current increases (and e.m.f. the same) so power produced by battery increases	B1
	both the above statements and so the ratio decreases	B1